AI.108

Digital Signal Processi

COURSE OVERVIEW

Representing and manipulating signals in discrete time: sampling, filtering, transforms, and spectral analysis.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: SIGNALS IN MACHINES

KEY TOPICS

- Continuous vs. discrete signals
- Sampling and the Nyquist theorem
- Quantization and digital audio
- DSP in robotics and AI

TEACHING FOCUS

Convert the analog world into numbers.

KEY TAKEAWAY Digital signals are sampled slices of reality.

2

SESSION

60 MINUTES

DISCRETE-TIME SYSTEMS

KEY TOPICS

- Difference equations
- Linear time-invariant (LTI) systems
- Impulse response and convolution
- System stability

TEACHING FOCUS

Model systems that process sequences.

KEY TAKEAWAY LTI systems are analyzable and composable.

3

SESSION

60 MINUTES

THE Z-TRANSFORM

KEY TOPICS

- z-transform definition
- Region of convergence
- Transfer functions in z
- Poles and zeros

TEACHING FOCUS

Analyze discrete systems in the z-domain.

KEY TAKEAWAY The z-domain is to discrete what s-domain is to continu

4

SESSION

60 MINUTES

DIGITAL FILTERING

KEY TOPICS

- FIR vs. IIR filters
- Filter design: windowing method
- Low-pass, high-pass, band-pass
- Filtering sensor data

TEACHING FOCUS

Remove noise and extract bands of interest.

KEY TAKEAWAY Filters shape a signal's frequency content.

5

SESSION

⏱ 60 MINUTES

SPECTRAL ANALYSIS

KEY TOPICS

- Discrete Fourier Transform (DFT)
- FFT algorithm
- Spectral leakage and windowing
- Frequency-domain interpretation

TEACHING FOCUS

Reveal the frequencies inside a signal.

▲ **KEY TAKEAWAY** The spectrum shows what a signal is made of.

DIGITAL SIGNAL PROCESSING

6

SESSION

60 MINUTES

DSP IN PERCEPTION & REVIEW

KEY TOPICS

- Audio and sensor signal processing
- Real-time DSP on embedded
- Course recap and assessment
- Toward machine vision and perception

TEACHING FOCUS

DSP cleans the raw signals robots rely on.

KEY TAKEAWAY Good perception starts with clean signals.